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Unit 4 : Session 2 : SLO 1 | Pure Functions with Examples

I. Match the Following

1. `map()` Applies a function to each element. Example: `list(map(str.upper, ["a", "b"]))` → `['A', 'B']`.
2. `filter()` Selects elements based on a condition. Example: `list(filter(lambda x: x > 0, [-1, 2, -3, 4]))` → `[2, 4]`.
3. `reduce()` Accumulates values to a single result. Example: `reduce(lambda a, b: a + b, [1, 2, 3, 4])` → `10`.
4. Pure Function Same output for the same input, with no side effects. Example: `def add(a, b): return a + b`.

II. Key Notes

- A pure function never mutates its arguments, global variables, or performs I/O → its result depends only on its inputs.
- `map()` and `filter()` return lazy iterators in Python 3; wrap them in `list()` to materialize the results.
- `reduce()` lives in the `functools` module: `from functools import reduce`.